

PERMANENT-ON-TOP BOUNDS IN ORDER FOUR

MINGCHANG LIU

ABSTRACT. For a Hermitian positive semidefinite matrix A , let $\Pi(A)$ denote its Schur power matrix. The all-ones vector is an eigenvector of $\Pi(A)$ with eigenvalue $\text{per}(A)$; the permanent-on-top assertion is that this eigenvalue is largest. We prove this assertion for every real symmetric positive semidefinite matrix of order four, with no rank restriction, and for every complex Hermitian positive semidefinite matrix of order four and rank at most two. For every complex Hermitian positive semidefinite matrix of order four, we prove $\lambda_{\max}(\Pi(A)) \leq 19 \text{per}(A)/18$, as well as the permanent-on-top assertion whenever an off-diagonal entry vanishes. We also bound each eigenvalue of the blocks corresponding to the irreducible S_4 representations $(2, 2)$ and $(2, 1, 1)$ by the permanent. Together with known order-five counterexamples, an explicit family in every order at least six gives a sharp cutoff at order four for the complex rank-two restriction. The proofs combine representation-theoretic reductions, exact rational polynomial certificates, and a diagonal-decrement identity. The factor $19/18$ is not claimed to be optimal, and the unrestricted complex Hermitian positive semidefinite case of order four remains open.

1. INTRODUCTION

For an $n \times n$ Hermitian positive semidefinite matrix $A = (a_{ij})$, index a matrix by the permutations of $[n]$ and put

$$(1) \quad \Pi(A)_{\sigma, \tau} = \prod_{j=1}^n a_{\sigma(j), \tau(j)} \quad (\sigma, \tau \in S_n).$$

This Schur power matrix is positive semidefinite and has the permanent $p = \text{per}(A)$ as a row-sum eigenvalue. The permanent-on-top (POT) conjecture, proposed by Soules, asserts that p is the largest eigenvalue in every order $[7, 3]$. Bapat and Sunder proved the assertion for $n \leq 3$ [1]. Counterexamples at order five disproved the general conjecture [6, 8]. The complex order-four case lies between these results and remains open [10, 9].

The principal results here separate the real and low-rank positive cases from the remaining complex obstruction.

2020 *Mathematics Subject Classification.* 15A15, 15A18, 05E10.

Key words and phrases. permanent, Schur power matrix, positive semidefinite matrix, symmetric group, exact polynomial certificate.

Georgia Institute of Technology, Atlanta, GA, USA.

Corresponding author: mliu416@gatech.edu.

ORCID: 0009-0005-3830-8742.

Theorem 1.1. *If A is a real symmetric positive semidefinite matrix of order four, or a complex Hermitian positive semidefinite matrix of order four with $\text{rank } A \leq 2$, then*

$$\lambda_{\max}(\Pi(A)) = \text{per}(A).$$

The real assertion includes every rank and every singular boundary. It does not extend to all real orders: Drury found a real rank-three counterexample at order sixteen [2]. The complex rank-two assertion has a sharp order boundary, stated below.

For a general complex Hermitian positive semidefinite matrix of order four, the irreducible S_4 sectors indexed by $(2, 2)$ and $(2, 1, 1)$ obey the permanent bound. In the remaining $(3, 1)$ sector we obtain a quantitative substitute.

Theorem 1.2. *Every Hermitian positive semidefinite matrix A of order four satisfies*

$$\text{per}(A) \leq \lambda_{\max}(\Pi(A)) \leq \frac{19}{18} \text{per}(A).$$

If $a_{ij} = 0$ for some $i \neq j$, then $\lambda_{\max}(\Pi(A)) = \text{per}(A)$.

The factor $19/18$ is an upper bound, not asserted to be optimal. The remaining $(3, 1)$ sector is represented by the cofactor-permanent matrix $F_A = (a_{ij} \text{per } A(i, j))$. Bapat and Sunder asked whether its eigenvalues are bounded by $\text{per}(A)$ in every order; Drury found a counterexample and asked which Fourier sectors retain the bound [1, 3]. Pioge and coauthors related this F_A question to a separate Hadamard-product inequality [5]. The operator bounds for the $(2, 2)$ and $(2, 1, 1)$ sectors are stronger than normalized character inequalities: the latter control traces, whereas permanent-on-top requires every eigenvalue to be controlled. Our proofs use associated matrices in permutation representations of S_4 , exact rational sum-of-squares certificates on Gram charts, and a diagonal-decrement identity to pass from rank three to rank four. Recent preprints by Siwei Zeng on generalized matrix functions in order four (arXiv:2608.21749v1) and by Yinjie Li on ordinary immanants through order fifteen (arXiv:2609.13412v1) concern character averages rather than full Fourier-block spectra.

The rank-two result at order four completes a restricted order classification. The negative direction uses known order-five examples and an explicit family in every order at least six.

Theorem 1.3. *For a fixed n , every complex Hermitian positive semidefinite matrix A of order n with $\text{rank } A \leq 2$ satisfies $\lambda_{\max}(\Pi(A)) = \text{per}(A)$ if and only if $n \leq 4$.*

We first set out the group-matrix reduction and the rank-decrement identity. Section 3 proves the rank-two result and its all-order cutoff; Section 4 proves the real order-four theorem. The complex sector bounds and remaining determinant appear in Section 5.

2. SCHUR POWERS AND RANK REDUCTION

The matrix in (1) is the Gram matrix of the tensors $v_{\sigma(1)} \otimes \cdots \otimes v_{\sigma(n)}$ when $A = (\langle v_i, v_j \rangle)$. In particular $\Pi(A) \succeq 0$. Its row sums are p , so $\lambda_{\max}(\Pi(A)) \geq p$. If a Gram vector vanishes, then $p = 0$ and $\Pi(A) = 0$; hence all statements below include this boundary case.

The proofs below use the irreducible Fourier blocks of the group matrix $\Pi(A)$. Their associated matrices are defined explicitly in the real and complex sections. This representation method follows [8, Section 2]; the coefficient conventions used here are also checked directly in the accompanying verification data. Under a nonsingular diagonal congruence $A \mapsto D^*AD$, both p and $\Pi(A)$ scale by $\prod_i |d_i|^2$. We use this to simplify Gram coordinates on dense sets, then restore the boundary by continuity.

Soules showed that a least-order real counterexample, if one exists, can be sought among singular matrices with further restrictions [7]. The following diagonal-decrement identity gives the rank extension used here. It is stated with a general factor because we use it both for equality and for the 19/18 bound.

Lemma 2.1 (Diagonal decrement). *Let $A \succeq 0$ have order n , let $t \geq 0$, and suppose $B = A - tE_{ii} \succeq 0$, where $E_{ii} = e_i e_i^*$ is the i th diagonal matrix unit. Write $A(i, i)$ for the principal submatrix obtained by deleting row and column i . For $\kappa \geq 1$ put $D_{\kappa, n}(A) = \kappa \operatorname{per}(A)I - \Pi(A)$. After ordering permutations by the position occupied by i ,*

$$(2) \quad D_{\kappa, n}(A) = D_{\kappa, n}(B) + t \bigoplus_{j=1}^n D_{\kappa, n-1}(A(i, i)).$$

If A is positive definite, $t = ((A^{-1})_{ii})^{-1}$ makes B positive semidefinite of rank $n - 1$.

Proof. Every term in the permanent and every Schur-power entry is affine in a_{ii} . An entry of $\Pi(A)$ uses a_{ii} exactly when its two permutations place i in the same position. In that block its coefficient is the corresponding entry of $\Pi(A(i, i))$; the coefficient in $\operatorname{per}(A)$ is $\operatorname{per}(A(i, i))$. This proves (2). For the last assertion, $A^{-1/2}BA^{-1/2} = I - t(A^{-1/2}e_i)(A^{-1/2}e_i)^*$ has one zero eigenvalue and all others equal to one. $\square$

3. RANK-TWO MATRICES

We first prove the permanent-on-top inequality for complex matrices of order four and rank at most two. The argument also identifies exactly where its finite certificate enters. We then give rank-two counterexamples in every order at least five. Throughout this section, $A = V^*V$, with columns $v_1, \dots, v_n \in \mathbb{C}^2$, $p = \operatorname{per} A$, and $\Pi(A)_{\sigma, \tau} = \prod_{j=1}^n a_{\sigma(j), \tau(j)}$.

An SSRN abstract by Zijian Zeng, posted in August 2026 and later withdrawn (<https://doi.org/10.2139/ssrn.7362322>), announced the real rank-two case at order four and partial complex rank-two block estimates.

It did not state the full complex rank-two inequality proved below; no full text was available for a proof-level comparison.

3.1. The order-four inequality. For $1 \leq i < j \leq 4$, put

$$\Delta_{ij} = \det(v_i, v_j), \quad \delta_{ij} = |\Delta_{ij}|^2,$$

and write $\{k, l\} = \{1, 2, 3, 4\} \setminus \{i, j\}$. In the standard orthonormal basis of $\text{Sym}^2 \mathbb{C}^2$, set

$$s_{kl} = \begin{pmatrix} (v_k)_1(v_l)_1 \\ ((v_k)_1(v_l)_2 + (v_k)_2(v_l)_1)/\sqrt{2} \\ (v_k)_2(v_l)_2 \end{pmatrix}, \quad B = 2 \sum_{i < j} \delta_{ij} s_{kl} s_{kl}^*,$$

$$q = 2 \sum_{\{\{i,j\}, \{k,l\}\}} \delta_{ij} \delta_{kl},$$

where the last sum runs over the three partitions into two pairs.

Lemma 3.1. *The spectrum of $\Pi(A)$, with multiplicities, is*

$$\{p\} \cup \text{spec}(B)^{\times 3} \cup \{q, q\} \cup \{0\}^{\times 12}.$$

In particular, $B \succeq 0$ and $q \geq 0$.

Proof. Let $w = v_1 \otimes \cdots \otimes v_4$ and let U_σ permute tensor positions. The operator with columns $U_\sigma w$ has Gram matrix permutation-similar to $\Pi(A)$. Its frame operator $\Omega = \sum_\sigma U_\sigma w w^* U_\sigma^*$ therefore has the same nonzero spectrum. Schur–Weyl duality decomposes $(\mathbb{C}^2)^{\otimes 4}$ into the (4), (3, 1), and (2, 2) components. Their Specht factors have dimensions 1, 3, 2, while their $SU(2)$ spin factors have dimensions 5, 3, 1.

To determine the normalization of the latter two blocks, let M be a matching of r disjoint pairs, leaving $s = 4 - 2r$ positions. The map T_M contracts each pair against the normalized singlet $(e_1 \otimes e_2 - e_2 \otimes e_1)/\sqrt{2}$ and orthogonally symmetrizes the unpaired positions. It is an $SU(2)$ -equivariant coisometry onto $\text{Sym}^s \mathbb{C}^2$, supported on the component $\lambda = (4 - r, r)$. There are $N = 4!/(2^r r! s!)$ such matchings. The sum $\sum_M T_M^* T_M$ commutes with $SU(2)$ and S_4 , since permuting tensor positions permutes the matchings and the singlet signs cancel. Irreducibility for $S_4 \times SU(2)$ therefore makes it a scalar multiple of the isotypic projector P_λ . Writing f^λ for the S_4 -dimension of λ , trace comparison gives

$$\sum_M T_M^* T_M = \frac{N}{f^\lambda} P_\lambda,$$

because each summand has trace $s + 1$ and $\text{tr } P_\lambda = f^\lambda(s + 1)$. Write the restriction of the permutation twirl to this isotypic component, with the spin space identified equivariantly and unitarily with $\text{Sym}^s \mathbb{C}^2$, as $I_{f^\lambda} \otimes X_\lambda$. On this component, $SU(2)$ -equivariance and Schur’s lemma identify each T_M with a unit functional on the Specht factor tensored with the identity on the spin factor; its coisometry property gives the unit normalization. Thus $T_M \Omega T_M^* = X_\lambda$ for each matching, and summing gives $N X_\lambda$. On the other

hand, permuting tensor positions permutes the matchings, up to signs that disappear in outer products. Expanding Ω therefore gives

$$NX_\lambda = 4! \sum_M (T_M w)(T_M w)^*.$$

For one paired singlet, $T_{\{i,j\}} w = (\Delta_{ij}/\sqrt{2}) s_{kl}$. For two paired singlets, its image is $\Delta_{ij}\Delta_{kl}/2$, up to sign. Substitution gives

$$r!s! \sum_M \left(\prod_{\{i,j\} \in M} \delta_{ij} \right) s_{[4] \setminus \cup M} s_{[4] \setminus \cup M}^*,$$

where s_R is the symmetric projection of the tensor product of the unpaired columns and $s_\emptyset = 1$. Indeed, the coefficient $(4!/N)2^{-r} = r!s!$. For $r = 1$ this is B , and for $r = 2$ it is q . The symmetric component has nonzero eigenvalue $4!\|P_{\text{sym}} w\|^2 = p$. These account for $1 + 9 + 2 = 12$ possible nonzero eigenvalues of the 24-dimensional Schur power. The remaining twelve are zero. $\square$

Theorem 3.2. *If $A \in \mathbb{C}^{4 \times 4}$ is Hermitian positive semidefinite and $\text{rank } A \leq 2$, then $\Pi(A) \preceq pI$.*

Proof. The assertion is immediate when a column of V is zero. Otherwise positive rescaling of the columns lets us assume $\|v_i\| = 1$. Let $n_i \in \mathbb{R}^3$ be the Bloch vector of v_i , so that $\delta_{ij} = (1 - n_i \cdot n_j)/2$. Put

$$E = \delta_{12}\delta_{34} + \delta_{13}\delta_{24} + \delta_{14}\delta_{23}, \quad k = \frac{1}{4} \left\| \sum_{i=1}^4 n_i \right\|^2.$$

Direct expansion of the permanent and the two blocks gives

$$(3) \quad p = 6k + 2E, \quad q = 2E, \quad \text{tr } B = 2(4 - k - E).$$

Thus $q \leq p$. To bound the trace of B , apply the normalized four-by-four Hadamard transform, with rows $(++++), (+-+-), (++--), (+-- +)$, to the matrix with columns $n_1, \dots, n_4$. Denote the transformed columns by u_0, u_1, u_2, u_3 ; their Gram matrix has the form

$$G = \begin{pmatrix} k & x & y & z \\ x & a & -z & -y \\ y & -z & b & -x \\ z & -y & -x & c \end{pmatrix} \succeq 0, \quad k + a + b + c = 4,$$

where a, b, c are the squared norms of u_1, u_2, u_3 , and x, y, z are their inner products with u_0 , respectively. An expansion of (3) yields

$$(4) \quad \begin{aligned} 2p - \text{tr } B &= 2k(k+2) + \frac{(a-b)^2 + (a-c)^2 + (b-c)^2}{8} \\ &\quad + \frac{3}{2}((ka - x^2) + (kb - y^2) + (kc - z^2)) \geq 0. \end{aligned}$$

Every term in the last parentheses is a principal minor of G .

It remains to prove $\det(pI - B) \geq 0$. On the dense set where $v_1 \neq 0$ and $\langle v_1, v_j \rangle \neq 0$ for $j = 2, 3, 4$, a common unitary, nonzero column scalings, and a residual phase bring V to

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & r & a + ib & c + id \end{pmatrix}, \quad r, a, b, c, d \in \mathbb{R}.$$

Under these column scalings, p and the eigenvalues of B are multiplied by a common factor $\gamma > 0$. Thus the trace defect scales by γ and $\det(pI - B)$ by γ^3 , preserving their signs. Write $z = (0, r, a + ib, c + id)$, $t = (r, a, b, c, d)$, and let $e_j = e_j(z)$ be the elementary symmetric polynomials. The permanent becomes $p = 24 + 6|e_1|^2 + 4|e_2|^2 + 6|e_3|^2$. With $g_{kl} = (1, z_k + z_l, z_k z_l)^T$, define

$$M(t) = p \operatorname{diag}(1, 2, 1) - 2 \sum_{i < j} |z_i - z_j|^2 g_{kl} g_{kl}^*.$$

This is a congruence of $pI - B$ by $\operatorname{diag}(1, \sqrt{2}, 1)$; hence $\det M = 2 \det(pI - B)$. The exact rational certificate supplied with the computational materials is the polynomial identity

$$(5) \quad \det M(t) = 103680 \sum_{h=1}^4 \left(\beta_h(t)^T Q_h \beta_h(t) + \beta_h(1t)^T Q_h \beta_h(1t) \right),$$

$$\iota(r, a, b, c, d) = (r, c, d, a, b).$$

Here the β_h are explicit rational polynomial vectors of lengths 69, 61, 66, 64, respectively, and each Q_h is an explicit positive-definite rational matrix. The verifier reconstructs p by its 24-permutation definition and checks every coefficient of (5) over $\mathbb{Q}$; it certifies $Q_h \succ 0$ by integer congruences to strictly diagonally dominant matrices with positive diagonal. Thus $\det(pI - B) \geq 0$ on this chart.

Since $B \succeq 0$ and $\operatorname{tr} B \leq 2p$, a hypothetical eigenvalue $\lambda_1(B) > p$ would leave the other two eigenvalues strictly below p , and would make $\det(pI - B) < 0$. Hence $B \preceq pI$. Lemma 3.1 proves the claim on the dense chart. The maps $V \mapsto p$ and $V \mapsto \Pi(V^*V)$ are continuous and the positive semidefinite cone is closed, so the claim holds also at every chart boundary. $\square$

3.2. The sharp order cutoff for rank two. The positive result at order four is the last universal rank-two case. The order-five counterexample below is the construction of Tran [8]; rank-two order-five counterexamples were first given by Shchesnovich [6].

Theorem 3.3. *For a positive integer n , every complex Hermitian positive semidefinite $n \times n$ matrix of rank at most two satisfies $\Pi(A) \preceq \operatorname{per}(A)I$ if and only if $n \leq 4$.*

Proof. The cases $n \leq 3$ follow from the order-three theorem of Bapat and Sunder [1], and $n = 4$ is Theorem 3.2. For $n = 5$, put $z = (i, -1, -i, 1, 0)$ and $A = (1 + 2\bar{z}_i z_j)_{i,j=1}^5$. This is a rank-two Gram matrix and $\operatorname{per} A = 504$.

Index a vector $y \in \mathbb{C}^{\binom{5}{2}}$ by the two-subsets in lexicographic order, and take

$$y = (-1 + i, 0, 1 + i, -2i, -1 - i, 0, 2, 1 - i, 2i, -2).$$

The ten-by-ten matrix $C_{S,T} = \text{per } A[S, T] \text{ per } A[S^c, T^c]$ satisfies $Cy = 512y$, as direct arithmetic with Gaussian integers verifies. If $c_\sigma = y_{\{\sigma(1), \sigma(2)\}}$, grouping the permutations τ by $\{\tau(1), \tau(2)\}$ shows that $\Pi(A)c = 512c$. Since $c \neq 0$, this violates the inequality.

For every $n \geq 6$ we use a single family. Let the six initial columns be the normalized vectors

$$e_0, e_1, \frac{e_0 + e_1}{\sqrt{2}}, \frac{e_0 - e_1}{\sqrt{2}}, \frac{e_0 + ie_1}{\sqrt{2}}, \frac{e_0 - ie_1}{\sqrt{2}}.$$

For $n = 6 + 2a$, append a copies of each pole e_0, e_1 . For $n = 7 + 2a$, append $a + 1$ copies of e_0 and a copies of e_1 , where $a \geq 0$. Both Gram matrices have rank two.

Here is an exact calculation of the violating sectors. For arbitrary columns $v_i = (\alpha_i, \beta_i)^T$, if $\prod_i (\alpha_i X + \beta_i Y) = \sum_{j=0}^n c_j X^{n-j} Y^j$, then projection onto the symmetric tensors gives

$$(6) \quad \text{per}(V^*V) = \sum_{j=0}^n j!(n-j)!|c_j|^2.$$

Let $g = q_0 I - i(q_1 \sigma_x + q_2 \sigma_y + q_3 \sigma_z) \in SU(2)$, where (q_0, q_1, q_2, q_3) is uniform on the unit sphere in $\mathbb{R}^4$. With $w = \bigotimes_i v_i$, set $f(g) = \langle w, g^{\otimes n} w \rangle$. The six initial vectors give

$$f_6(g) = (q_0^2 + q_1^2)(q_0^2 + q_2^2)(q_0^2 + q_3^2).$$

Schur–Weyl duality decomposes the tensor frame operator into blocks $X_j \otimes I_{f(n-j,j)}$. The spin space of X_j has dimension $n - 2j + 1$. The Gram/frame duality in the proof of Lemma 3.1 applies in every order. Thus a Rayleigh quotient of any block gives a lower bound for $\lambda_{\max}(\Pi(A))$. If $n = 2m$, its lowest-spin block X_m is scalar and character orthogonality gives

$$(7) \quad X_m = \frac{n!}{C_m} \int_{SU(2)} f(g) dg, \quad C_m = \frac{1}{m+1} \binom{2m}{m}.$$

If $n = 2m + 1$, then X_m acts on $\mathbb{C}^2$, and Schur orthogonality of the defining representation gives, for unit $\eta \in \mathbb{C}^2$,

$$(8) \quad \langle \eta, X_m \eta \rangle = \frac{2n!}{f^{(m+1,m)}} \int_{SU(2)} \overline{\langle \eta, g\eta \rangle} f(g) dg, \quad f^{(m+1,m)} = \frac{2(2m+1)!}{m!(m+2)!}.$$

For completeness, the factor 2 in (8) is the dimension of the defining representation: averaging $2\overline{\langle \eta, g\eta \rangle} g^{\otimes n}$ projects onto $|\eta\rangle\langle\eta|$ in the spin-one-half summand and annihilates every other summand. Taking its trace against the tensor frame operator proves the formula. The same argument with the trivial character proves (7).

Writing $P_j = q_0^2 + q_j^2$, direct integration on the sphere gives

$$(9) \quad I(b) := \int_{SU(2)} P_3^{b+1} P_1 P_2 dg = \frac{3b+7}{8(b+2)(b+3)}, \quad b \geq 0.$$

Indeed, $u = P_3$ is uniform on $[0, 1]$. Conditional on u , the two coordinate pairs have independent uniform angular variables. Their angular average of $P_1 P_2$ is $\frac{3}{8}u^2 + \frac{1}{2}u(1-u) + \frac{1}{8}(1-u)^2$; integrating this against u^{b+1} yields (9).

For $n = 6 + 2a$, (6) gives $p = (a+1)!(a+5)!/8$, while (7)–(9) give

$$\frac{X_{a+3}}{p} = \frac{3a+7}{a+5} > 1.$$

For $n = 7 + 2a$, the same permanent formula gives

$$p = \frac{(a+1)!(a+5)!(a+4)}{8}.$$

Taking $\eta = e_0$ in (8), the extra factor $\overline{\langle e_0, ge_0 \rangle}$ combines with the one unpaired pole to make P_3^{a+2} . Consequently

$$\frac{\langle e_0, X_{a+3}e_0 \rangle}{p} = \frac{(a+2)(3a+10)}{(a+4)^2} > 1,$$

since the numerator exceeds the denominator by $2a^2 + 8a + 4$. A Rayleigh quotient in either case exceeds the permanent, completing the proof. $\square$

4. THE REAL ORDER-FOUR THEOREM

We prove the real case first on the singular locus. Two coordinate charts are useful: one gives compact expressions for eight of the required inequalities, while the other gives a sum-of-squares identity for the remaining hook determinant. The positive-definite case then follows by decreasing one diagonal entry.

Theorem 4.1. *For every real symmetric positive-semidefinite matrix A of order four,*

$$\lambda_{\max}(\Pi(A)) = \text{per } A.$$

4.1. The three nontrivial sectors. Write $p = \text{per } A$ and define

$$\begin{aligned} F_{ij} &= a_{ij} \text{ per } A(i, j), \\ G_{ij} &= a_{ij}(-1)^{i+j} \det A(i, j), \\ C_{I,J} &= \text{per } A[I, J] \text{ per } A[I^c, J^c] \quad (|I| = |J| = 2). \end{aligned}$$

Here $A(i, j)$ is obtained by deleting row i and column j , whereas $A[I, J]$ denotes a submatrix. Order the two-element subsets as 12, 13, 14, 23, 24, 34,

and put

$$Q = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix}, \quad R = \begin{pmatrix} 1 & 1 \\ -1 & 0 \\ 0 & -1 \\ 0 & -1 \\ -1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The defect matrices used below are

$$(10) \quad \begin{aligned} K_{31} &= Q^\top(pI_4 - F)Q, & K_{211} &= Q^\top(pI_4 - G)Q, \\ K_{22} &= R^\top(pI_6 - C)R. \end{aligned}$$

The columns of Q span $\mathbf{1}^\perp$. The columns of R span the kernel of the vertex-subset incidence matrix, whose (i, I) entry is 1 when $i \in I$ and 0 otherwise.

For completeness, these matrices identify all the sectors needed here. Set $\phi(\sigma) = \prod_i a_{i, \sigma(i)}$. The natural permutation representation of S_4 decomposes as $(4) \oplus (3, 1)$, its sign twist as $(1, 1, 1, 1) \oplus (2, 1, 1)$, and the action on two-element subsets as $(4) \oplus (3, 1) \oplus (2, 2)$. Summing $\phi(\sigma)$ over $\sigma(i) = j$, with or without $\text{sgn } \sigma$, gives F_{ij} and G_{ij} , respectively, up to transpose; summing over $\sigma(I) = J$ gives $C_{I, J}$. These are consequently the associated operators in the three permutation representations. Restriction to the indicated invariant subspaces gives the $(3, 1)$, $(2, 1, 1)$ and $(2, 2)$ blocks. The bases Q and R need not be orthonormal: congruence tests positivity of their quadratic forms. The two remaining blocks are the scalars p and $\det A$. These five irreducible representations have dimensions 1, 3, 2, 3, 1, whose squares sum to 24, so they exhaust the Schur power. In particular, when $\text{rank } A \leq 3$, the sign block is zero and

$$(11) \quad pI_{24} - \Pi(A) \succeq 0 \iff K_{31}, K_{211}, K_{22} \succeq 0.$$

Here $p \geq 0$ follows, for example, because $\Pi(A)$ is a Gram matrix and its row sum is p .

4.2. Exact polynomial inequalities. For a symmetric 3×3 matrix H , let $e_j(H)$ be the j th elementary symmetric function of its eigenvalues. Thus $e_1(H)$ is its trace, $e_2(H)$ is the sum of its three principal minors of order two, and $e_3(H) = \det H$. We use the following two real Gram charts:

$$(12) \quad V_0 = \begin{pmatrix} 1 & x & y & u \\ 0 & 1 & z & v \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad V_1 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & x & y & u \\ 0 & 0 & z & v \end{pmatrix}.$$

All five parameters are unrestricted real numbers.

Lemma 4.2. *For $A = V_0^\top V_0$, each of the first eight polynomials in Table 1 is nonnegative. For $A = V_1^\top V_1$, the polynomial $\det K_{31}$ is nonnegative.*

TABLE 1. Exact sum-of-squares identities for the real case. The last column gives the dimensions of the four rational Gram matrices. The first eight rows use V_0 and the last uses V_1 .

Polynomial	Degree	Nonzero terms	Gram dimensions
$(K_{22})_{11}$	6	32	4, 4, 4, 4
$(K_{22})_{22}$	6	32	4, 4, 4, 4
$\det K_{22}$	12	272	12, 11, 11, 17
$e_1(K_{211})$	6	32	5, 4, 4, 5
$e_2(K_{211})$	12	297	32, 24, 24, 28
$\det K_{211}$	18	1401	87, 96, 87, 101
$e_1(K_{31})$	6	32	5, 4, 4, 5
$e_2(K_{31})$	12	290	19, 19, 21, 24
$\det K_{31}$	16	1240	45, 40, 42, 39

Proof. The exact data supplied with the paper express each polynomial f as

$$(13) \quad f = s \sum_{h=0}^3 \beta_h^\top M_h \beta_h, \quad s \in \mathbb{Z}_{>0}, \quad M_h \in \mathbb{Q}^{d_h \times d_h},$$

where the entries of β_h are rational polynomials and every M_h is positive definite. Both assertions can be checked using rational arithmetic: expanding the right side gives exactly the coefficients obtained from (10), and elimination without pivoting gives strictly positive diagonal entries in an $LDL^\top$ factorization of every M_h . The dimensions and supports are recorded in Table 1. The expansion checks all coefficients, including those required to vanish; it makes no numerical positivity test.

For the last identity, $s = 304128$; the four rational Gram matrices and their positive denominators are specified in the companion data. Its left side has constant coefficient 55296. Thus every right side in (13) is nonnegative for all real parameter values, proving the stated inequalities. $\square$

4.3. Coverage of the singular locus. Let $V = (v_1, v_2, v_3, v_4)$ be any real 3×4 matrix. A common orthogonal transformation does not change $V^\top V$. If its columns are multiplied by nonzero real numbers d_i , then p, F, G, C and $\Pi(A)$ are all multiplied by

$$(14) \quad \gamma = \prod_{i=1}^4 d_i^2 > 0.$$

Consequently the three defect matrices are multiplied by γ , and their scalar inequalities retain their signs.

On the dense open set

$$\det(v_1, v_2, v_3) \det(v_1, v_2, v_4) \neq 0,$$

Gram-Schmidt puts the first three columns in upper triangular form with nonzero diagonal. Nonzero column scalings make these diagonal entries and

the last coordinate of the fourth column equal to one. The resulting matrix is V_0 . Lemma 4.2 and (14) therefore give its first eight inequalities on a dense set of real factors. Since these are polynomial inequalities in the entries of V , they hold for every real factor by continuity.

For V_1 , use instead the dense open conditions

$$v_1^\top v_j \neq 0 \quad (1 \leq j \leq 4), \quad \|v_1\|^2 \|v_2\|^2 - (v_1^\top v_2)^2 \neq 0.$$

Send v_1 to the first coordinate axis and scale each column so that its first coordinate is one. A residual orthogonal transformation of the last two coordinates sends the nonzero residual of the second column to the second coordinate axis. This gives V_1 . The determinant inequality in Lemma 4.2 therefore also holds on a dense set and hence everywhere.

The three principal-minor inequalities now imply $K_{22} \succeq 0$. For each of $H = K_{31}, K_{211}$, all three coefficients $e_j(H)$ are nonnegative. Indeed,

$$\det(tI_3 + H) = t^3 + e_1(H)t^2 + e_2(H)t + e_3(H) > 0 \quad (t > 0).$$

A negative eigenvalue of H would give a positive root of this polynomial. Thus $H \succeq 0$, including on its singular boundary. Equation (11) proves Theorem 4.1 for rank at most three.

4.4. Passage to positive-definite matrices. Write $D_n(A) = D_{1,n}(A)$ in the notation of Lemma 2.1. For a positive-definite A , that lemma provides $B = A - tE_{ii} \succeq 0$ of rank $n - 1$ and expresses $D_n(A)$ as $D_n(B)$ plus n copies of the lower-order defect.

We first obtain the required order-three assertion without using it in the singular order-four proof. If C is a real positive-semidefinite matrix of order three and rank at most two, then $C \oplus [1]$ has order four and rank at most three. The result just proved applies to this matrix. Its Schur power has $\Pi(C)$ as a principal submatrix, and its permanent equals $\text{per } C$. Hence $D_3(C) \succeq 0$. For order two,

$$\Pi(A) = \begin{pmatrix} a_{11}a_{22} & a_{12}^2 \\ a_{12}^2 & a_{11}a_{22} \end{pmatrix},$$

so its largest eigenvalue is its row sum. Applying Lemma 2.1 at order three proves $D_3(C) \succeq 0$ also when C is positive definite. A final application at order four proves $D_4(A) \succeq 0$ for positive-definite real A . The row-sum eigenvalue of $\Pi(A)$ is $\text{per } A$, so the inequality is an equality for its largest eigenvalue. This completes the proof of Theorem 4.1.

5. COMPLEX MATRICES OF ORDER FOUR

We prove the permanent bound for two of the three nontrivial Fourier sectors of S_4 , and then treat a class on which the remaining sector also satisfies the bound. Throughout this section, A is a complex Hermitian positive semidefinite matrix of order four and $p = \text{per } A$. Write $X_\lambda(A)$ for an irreducible Fourier block of the Schur power corresponding to the partition

λ . Equivalent choices of unitary representation give the same spectrum. For the irreducible character χ^λ of S_4 , write

$$d_\lambda(A) = \sum_{\sigma \in S_4} \chi^\lambda(\sigma) \prod_{i=1}^4 a_{i, \sigma(i)}.$$

Theorem 5.1. *For every complex Hermitian positive semidefinite matrix A of order four,*

$$X_{(2,2)}(A) \preceq pI_2, \quad X_{(2,1,1)}(A) \preceq pI_3.$$

If $\text{rank } A \leq 3$, the unnormalized immanant satisfies the sharp bound

$$(15) \quad d_{(2,1,1)}(A) \leq \frac{8}{5}p.$$

The rank restriction in (15) is essential. For $A = I_4$ the immanant equals 3, whereas $p = 1$. The operator inequalities in the theorem have no rank restriction.

5.1. Associated matrices and the Gram chart. Let $P_0 = 12 \mid 34$, $P_1 = 13 \mid 24$, and $P_2 = 14 \mid 23$ be the three pair partitions of $\{1, 2, 3, 4\}$. For $g \in S_4$, put $\phi_A(g) = \prod_i a_{i, g(i)}$, and define

$$H_{ij} = \sum_{g(P_j)=P_i} \phi_A(g), \quad 0 \leq i, j \leq 2,$$

$$F_{ij} = a_{ij} \text{ per } A(i, j), \quad G_{ij} = a_{ij} (-1)^{i+j} \det A(i, j), \quad 1 \leq i, j \leq 4.$$

Here $A(i, j)$ is obtained by deleting row i and column j . These matrices are Hermitian, with

$$H\mathbf{1} = p\mathbf{1}, \quad F\mathbf{1} = p\mathbf{1}, \quad G\mathbf{1} = (\det A)\mathbf{1}.$$

On the orthogonal complements of the constant vectors, H , F , and G realize the $(2, 2)$, $(3, 1)$, and $(2, 1, 1)$ blocks, respectively, up to transpose and unitary equivalence. Thus a bound on one associated matrix bounds every copy of its corresponding block in the Schur power.

The cofactor formulas follow directly from the indexing in (1). For $y \in \mathbb{C}^4$, put $c_\sigma = y_{\sigma(1)}$ or $c_\sigma = \text{sgn}(\sigma)y_{\sigma(1)}$, respectively. Grouping the sum by $g = \tau\sigma^{-1}$ gives

$$(16) \quad c^* \Pi(A) c = 6y^* F y \quad \text{or} \quad 6y^* G y, \quad \|c\|^2 = 6\|y\|^2.$$

More explicitly, the lift $L : \mathbb{C}^4 \rightarrow \mathbb{C}^{S_4}$ defined by $(Ly)_\sigma = y_{\sigma(1)}$ satisfies

$$L^* L = 6I_4, \quad \Pi(A)L = LF.$$

Indeed, grouping the sum by $\tau(1) = j$ gives

$$(\Pi(A)Ly)_\sigma = \sum_j a_{\sigma(1), j} \text{ per } A(\sigma(1), j) y_j = (LFy)_\sigma.$$

Thus the image of L is invariant and the restriction is unitarily equivalent to F . The sign-twisted lift gives the corresponding relation with G .

For the pairing representation, fixing P_j gives the analogous realization on functions of $\sigma(P_j)$. The permutation representations on four points and on three pair partitions decompose as $(4) \oplus (3, 1)$ and $(4) \oplus (2, 2)$; twisting the first by the sign character gives $(1, 1, 1, 1) \oplus (2, 1, 1)$.

For rank at most three we use

$$(17) \quad V(t) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & x & a + ib & c + id \\ 0 & 0 & z & e + if \end{pmatrix}, \quad A(t) = V(t)^* V(t).$$

Here $t = (x, a, b, z, c, d, e, f) \in \mathbb{R}^8$. There are no sign restrictions on the parameters. To see why this chart suffices, start with a 3×4 Gram factor whose first column has nonzero inner product with every column and whose first two columns are independent. These conditions define an open dense subset of the space of factors. A common unitary transformation and nonzero column scalings set the first row to $(1, 1, 1, 1)$. A unitary transformation of the last two coordinates makes the residual of the second column equal to $(x, 0)$ with x real. A phase in the third coordinate then makes the third entry of the third column real. This gives (17).

For a nonsingular diagonal matrix D , the substitution $A \mapsto D^* A D$ multiplies each of p , $\Pi(A)$, H , F , and G by $\prod_i |D_{ii}|^2 > 0$. The inequalities below therefore transfer to the unitary and scaling orbit of the chart. Continuity in the original Gram factor extends them to every rank-at-most-three matrix. This argument includes vanishing inner products, dependent columns, and lower ranks.

Set

$$Q_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}, \quad Q_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & -1 \end{pmatrix},$$

and write

$$K_{22} = Q_2^*(pI_3 - H)Q_2, \quad K_{31} = Q_3^*(pI_4 - F)Q_3.$$

These are compressions in fixed nonorthonormal bases. Their characteristic coefficients are not the characteristic coefficients of the Fourier blocks. For a Hermitian matrix B , $e_j(B)$ denotes the j th elementary symmetric function of its eigenvalues.

5.2. Polynomial identities. We use exact identities for the targets in Table 2. Each has the form

$$(18) \quad P(t) = s \sum_{h=0}^7 \beta_h(t)^\top S_h \beta_h(t), \quad S_h \in \mathbb{Q}^{m_h \times m_h}, \quad S_h \succ 0,$$

where s is the indicated positive rational number and the entries of β_h are rational polynomials. The full bases and rational matrices are supplied with the exact verification data. The identities are checked by expanding both sides over $\mathbb{Q}$ and comparing every coefficient; positive definiteness of

S_h is checked by rational LDL^T elimination. Thus the identities imply nonnegativity on all of $\mathbb{R}^8$, including the singular points of the chart.

For the cofactor targets, put $T = \text{tr } G$ and $E = e_2(G)$.

TABLE 2. Exact polynomial identities used for the complex sectors. The last column gives $\sum_h m_h$, the number of positive rational pivots.

Target P	s	Degree	Pivots
$\text{tr } K_{22}$	96	6	40
$\det K_{22}$	3240	12	206
$\text{tr } K_{31}$	144	6	40
$e_2(K_{31})$	7968	12	354
$8p - 5T$	192	6	36
$3p^2 - 2pT - T^2 + 4E$	3024	12	340

Proof of Theorem 5.1 at rank at most three. The first two identities show that K_{22} has nonnegative trace and determinant. Hence $K_{22} \succeq 0$. Since the columns of Q_2 span $\mathbf{1}^\perp$ and $H\mathbf{1} = p\mathbf{1}$, we obtain $H \preceq pI_3$.

For the cofactor sector, let $A = V^*V$ and set $k_i = (-1)^{i-1} \det V_{\hat{i}}$, where $V_{\hat{i}}$ omits column i . Laplace expansion and the Cauchy–Binet formula give

$$Vk = 0, \quad (-1)^{i+j} \det A(i, j) = \bar{k}_i k_j.$$

Consequently

$$(19) \quad G = \text{diag}(\bar{k})A \text{diag}(k) \succeq 0, \quad G\mathbf{1} = 0.$$

Let the eigenvalues of G on $\mathbf{1}^\perp$ be $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 0$. The last two identities in the table give

$$T \leq \frac{8}{5}p, \quad 3p^2 - 2pT - T^2 + 4E \geq 0.$$

Furthermore,

$$4(T^2 - 3E) - (3\lambda_1 - T)^2 = 3(\lambda_2 - \lambda_3)^2 \geq 0.$$

Because $3\lambda_1 - T \geq 0$, this implies

$$\lambda_1 \leq \frac{T + 2\sqrt{T^2 - 3E}}{3}.$$

The bound on T ensures $3p - T \geq 0$, and

$$(3p - T)^2 - 4(T^2 - 3E) = 3(3p^2 - 2pT - T^2 + 4E) \geq 0.$$

It follows that $\lambda_1 \leq p$. The argument uses no division by p ; if $p = 0$, the trace bound and (19) give $G = 0$ directly. Since $\det A = 0$, $T = d_{(2,1,1)}(A)$, proving the trace assertion. The chart and continuity argument transfers all these inequalities to every rank-at-most-three matrix.

For sharpness, take $A = I_4 - J_4/4$. Then A is a rank-three orthogonal projection, $p = 15/32$, $G = A/4$, and $T = 3/4$. Hence $T/p = 8/5$. $\square$

5.3. Passage to full rank. Lemma 2.1 gives the rank reduction used here and below. For positive definite A of order four, it writes the full defect as the defect of a rank-three matrix plus four lower-order defects. The order-three permanent-on-top theorem of Bapat and Sunder [1] makes the added operator in (2) positive semidefinite. Restricting to either of the two sectors in Theorem 5.1 therefore proves its full-rank assertions. This use of the order-three theorem does not extend the trace bound (15) to rank four.

5.4. A zero off-diagonal entry.

Theorem 5.2. *If A has a zero off-diagonal entry, then $\lambda_{\max}(\Pi(A)) = \text{per } A$.*

Proof. After relabelling, suppose $a_{12} = 0$. First assume $\text{rank } A \leq 3$. If a Gram vector v_1 or v_2 is zero, then $p = \Pi(A) = 0$. Otherwise those vectors are nonzero and orthogonal. A common unitary and nonzero column scalings make them e_1, e_2 . On the relatively dense subset where the third coordinates of v_3, v_4 do not vanish, further column scalings and coordinate phases give the chart

$$(20) \quad V_0 = \begin{pmatrix} 1 & 0 & a & c + id \\ 0 & 1 & b & e + if \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad (a, b, c, d, e, f) \in \mathbb{R}^6.$$

There is an exact identity

$$(21) \quad \det K_{31}(V_0^* V_0) = 10240 \sum_{h=0}^7 \gamma_h^T R_h \gamma_h, \quad R_h \succ 0,$$

with rational polynomial bases of lengths 24, 26, 20, 21, 20, 22, 22, 21 and rational matrices R_h . Expansion verifies all 625 coefficient positions of this degree-twelve polynomial, including its zero coefficients; rational elimination gives 176 positive pivots.

The previous identities give $e_1(K_{31}), e_2(K_{31}) \geq 0$ for every matrix of rank at most three. Together with (21), they imply $K_{31} \succeq 0$. In fact, if a Hermitian matrix of order three with these three nonnegative characteristic coefficients had an eigenvalue $-s < 0$, then

$$0 = \det(sI + K_{31}) = s^3 + e_1(K_{31})s^2 + e_2(K_{31})s + \det K_{31} > 0,$$

a contradiction. Thus the $(3, 1)$ block is bounded by p . Theorem 5.1 bounds the other two nontrivial blocks; the sign block is $\det A = 0$ and the trivial block is p .

The scaling argument transfers the result along the orbit of (20). Perturbing only the third coordinates of v_3, v_4 and passing to the limit covers its boundary while preserving $a_{12} = 0$. This proves the theorem at rank at most three. Finally, the rank reduction (2) changes only a diagonal entry, so it preserves $a_{12} = 0$ and proves the full-rank case. $\square$

5.5. A universal quantitative bound.

Theorem 5.3. *For every complex Hermitian positive semidefinite matrix A of order four,*

$$(22) \quad \text{per } A \leq \lambda_{\max}(\Pi(A)) \leq \frac{19}{18} \text{per } A.$$

Proof. On the chart (17), let $y = (q_0 + iq_3, q_1 + iq_4, q_2 + iq_5)^\top$, with $q \in \mathbb{R}^6$. The polynomial

$$\Theta(t, q) = y^* Q_3^* (19pI_4 - 18F) Q_3 y$$

has the exact identity

$$(23) \quad \Theta(t, q) = 912 \sum_{h=0}^7 \eta_h(t, q)^\top U_h \eta_h(t, q), \quad U_h \succ 0.$$

The vectors η_h consist of monomials. Their eight lengths alternate between 33 and 27. The matrices are rational. Exact expansion verifies 1392 coefficient positions, and rational elimination gives 240 positive pivots. The identity has no multiplier or restriction on (t, q) .

It follows that $19pI_4 - 18F$ is positive semidefinite on $\mathbf{1}^\perp$. On the constant line it acts by $p \geq 0$, so $F \preceq (19/18)pI_4$. The chart argument proves this hook bound at every rank at most three. Theorem 5.1 bounds the other nontrivial sectors by p , and the sign block vanishes. Thus the upper bound in (22) holds at rank at most three. Applying (2) with $\kappa = 19/18$ proves it at rank four: for an order-three matrix $C \succeq 0$,

$$D_{\kappa,3}(C) = D_{1,3}(C) + (\kappa - 1) \text{per}(C)I \succeq 0$$

by the order-three theorem [1]. The lower bound follows because p is a row-sum eigenvalue of $\Pi(A)$. $\square$

5.6. The remaining determinant. The preceding results isolate the unresolved part of complex order-four POT. In fact, they give a pointwise relation with the Bapat–Sunder matrix $F_A = F$ [1, 3]. For every $A \succeq 0$ of order four,

$$\lambda_{\max}(\Pi(A)) = \lambda_{\max}(F_A).$$

Indeed, the lift in (16) identifies the spectrum of F_A with $\{p\} \cup \text{spec } X_{(3,1)}(A)$. Theorem 5.1 bounds the $(2, 2)$ and $(2, 1, 1)$ blocks by p . The sign block has eigenvalue zero if $\text{rank } A \leq 3$; at full rank, apply Lemma 2.1 to its sign vector. The rank-three defect is $\text{per}(B)$ on this vector, and the added order-three defects are positive semidefinite by [1]. Thus the sign eigenvalue is also at most p . As p is an eigenvalue of F_A , no other block can exceed $\lambda_{\max}(F_A)$. Consequently the still-open complex order-four POT assertion is equivalent to $F_A \preceq pI_4$ in this order; the bound in Theorem 5.3 also applies to $\lambda_{\max}(F_A)$.

On the rank-at-most-three chart (17), the unresolved inequality is exactly

$$(24) \quad \det(Q_3^* (pI_4 - F) Q_3) \geq 0 \quad \text{for every } t \in \mathbb{R}^8.$$

The determinant has degree sixteen and 11704 nonzero integer coefficients. Its first two characteristic coefficients are nonnegative by Table 2. Hence

(24) would imply $K_{31} \geq 0$ by the argument used for Theorem 5.2. The chart, the other two sector bounds, and the rank reduction would then give full complex order-four POT. Conversely, POT implies (24) by compression. The determinant inequality is not established here.

6. CONCLUDING REMARKS

For complex matrices of order four, define

$$C_4 = \sup_{\substack{A \succeq 0 \\ \text{per}(A) > 0}} \frac{\lambda_{\max}(\Pi(A))}{\text{per}(A)}.$$

Theorem 1.2 gives $1 \leq C_4 \leq 19/18$. Whether $C_4 = 1$ is the remaining permanent-on-top question at this order. The bounds proved here for the other sectors and the first two characteristic coefficients reduce it to the determinant in (24); Lemma 2.1 then handles full rank. An exact negative value would instead give a counterexample.

The real rank-two problem at larger orders poses a different question: does the permanent bound every Schur-power eigenvalue of a real Gram matrix of rank at most two? Drury’s real counterexample has rank three [2] and does not decide this restriction.

STATEMENTS AND DECLARATIONS

Funding. The author received no funding for this work.

Competing interests. The author declares no competing interests.

Author contributions. Mingchang Liu is the sole author and takes responsibility for this work. The use of AI models is described below.

Data availability. The exact rational certificates and verification programs are available in release v1 of the companion repository [4]. The repository’s README describes the contents and gives instructions for reproducing the checks. The analytic steps from these identities to the stated matrix classes are given in the proofs.

AI statement. This manuscript grew out of continuous human–AI work on the order-four permanent-on-top problem. The author and AI models, including OpenAI’s GPT-6 Astra Pro and Anthropic’s Fable 5.1, developed and checked arguments and exact certificates together. The models also assisted with literature research and manuscript preparation. The author takes full responsibility for the mathematical results and the content of this manuscript.

REFERENCES

1. R. B. Bapat and V. S. Sunder, *An extremal property of the permanent and the determinant*, Linear Algebra Appl. **76** (1986), 153–163, [https://doi.org/10.1016/0024-3795\(86\)90220-X](https://doi.org/10.1016/0024-3795(86)90220-X).
2. Stephen Drury, *A real counterexample to two inequalities involving permanents*, Math. Inequal. Appl. **20** (2017), 349–352, <https://doi.org/10.7153/mia-20-23>.
3. ———, *A counterexample to a question of Bapat & Sunder*, Math. Inequal. Appl. **21** (2018), 517–520, <https://doi.org/10.7153/mia-2018-21-37>.
4. Mingchang Liu, *Permanent-on-top bounds in order four: exact certificates and verification programs*, <https://github.com/michaeliu4/permanent-on-top-order-four/releases/tag/v1>, 2026, Version v1, software and certificate data.
5. Léo Pioge, Kamil K. Pietrasz, Benoit Seron, Leonardo Novo, and Nicolas J. Cerf, *A logical implication between two conjectures on matrix permanents*, Linear Algebra Appl. **725** (2025), 309–318, <https://doi.org/10.1016/j.laa.2025.07.011>.
6. V. S. Shchesnovich, *The permanent-on-top conjecture is false*, Linear Algebra Appl. **490** (2016), 196–201, <https://doi.org/10.1016/j.laa.2015.10.034>.
7. George W. Soules, *An approach to the permanental-dominance conjecture*, Linear Algebra Appl. **201** (1994), 211–229, [https://doi.org/10.1016/0024-3795\(94\)90117-1](https://doi.org/10.1016/0024-3795(94)90117-1).
8. Hoang Anh Tran, *A simple counterexample for the permanent-on-top conjecture*, Math. Inequal. Appl. **25** (2022), 1–16, <https://doi.org/10.7153/mia-2022-25-01>.
9. Ian M. Wanless, *Lieb’s permanental dominance conjecture*, The Physics and Mathematics of Elliott Lieb: The 90th Anniversary (Rupert L. Frank, Ari Laptev, Mathieu Lewin, and Robert Seiringer, eds.), vol. 2, EMS Press, 2022, <https://doi.org/10.4171/90-2/48>, pp. 501–516.
10. Fuzhen Zhang, *An update on a few permanent conjectures*, Special Matrices **4** (2016), 305–316, <https://doi.org/10.1515/spma-2016-0030>.